$$\begin{aligned}
&\int_{\frac{X-F}{\tau}}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right)(\tau x + F - X)dx \\
&= \tau \int_{\frac{X-F}{\tau}}^{\infty} \frac{1}{\sqrt{2\pi}} x \exp\left(-\frac{x^2}{2}\right)dx \\
&\quad +(F-X)\int_{\frac{X-F}{\tau}}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right)dx. \qquad \text{(E.2)}
\end{aligned}$$

The first integral on the right-hand side of Equation E.2 is evaluated using integration by parts. The second integral on the right-hand side of Equation E.2 is equal to one less the cumulative normal distribution function evaluated at $(X-F)/\tau$. We have, then, that Equation E.2 is equivalent to

$$\begin{aligned}
&\frac{\tau}{\sqrt{2\pi}}\left[-\exp\left(-\frac{x^2}{2}\right)\right]_{\frac{X-F}{\tau}}^{\infty} + (F-X)\left[1-N\left(\frac{X-F}{\tau}\right)\right] \\
&= \frac{\tau}{\sqrt{2\pi}}\exp\left[-\frac{(X-F)^2}{2\tau^2}\right] + (F-X)\left[1-N\left(\frac{X-F}{\tau}\right)\right], \qquad \text{(E.3)}
\end{aligned}$$

where $N(\cdot)$ denotes the cumulative normal distribution function, that is,

$$N(d) = \int_{-\infty}^{d} \frac{1}{\sqrt{2\pi}} e^{-x^2/2}dx. \qquad \text{(E.4)}$$

Thus, the full value of the T into n payer swaption struck at X is

$$A\sigma_N\sqrt{T}\left[\frac{1}{\sqrt{2\pi}}e^{-d^2/2} + dN(d)\right], \qquad \text{(E.5)}$$

where

$$d = \frac{F-X}{\sigma_N\sqrt{T}} \qquad \text{(E.6)}$$

and

$$A = \sum_{i=1}^{n} \frac{1}{[1+zc(t_i)]^{t_i}} \qquad \text{(E.7)}$$

is the annuity factor (where $t_i = T + i$, for i = 1, 2, $\ldots, n$).

Using swaption parity, it follows immediately that the value of a T into n receiver swaption struck at X is

$$A\sigma_N\sqrt{T}\left[\frac{1}{\sqrt{2\pi}}e^{-d^2/2} - dN(-d)\right], \qquad \text{(E.8)}$$

with d and A as defined in Equations E.6 and E.7, respectively.

The value of a swaption straddle struck at-the-money forward is found by setting $X = F$ in each of Equations E.5 and E.8, then adding the two together. Thus, the value of a T into n swaption straddle struck at-the-money forward is given by

$$\sqrt{T} \times \sqrt{\frac{2}{\pi}} \times \sigma_N \times A. \tag{E.9}$$

E.1 The Relationship Between σ_{LN} and σ_N for Swaptions that Are Struck At-the-Money Forward

Under the assumption of lognormality, the value of a payer swaption and a receiver swaption are given by Black's formula:

$$A \times [FN(d_1) - XN(d_2)] \tag{E.10}$$

and

$$A \times [XN(-d_2) - FN(-d_1)], \tag{E.11}$$

respectively, where

$$d_1 = \frac{\ln(F/X) + \sigma_{LN}^2 T/2}{\sigma_{LN}\sqrt{T}} \tag{E.12}$$

and

$$d_2 = d_1 - \sigma_{LN}\sqrt{T}, \tag{E.13}$$

where σ_{LN} denotes lognormal vol.

Consider now the value of a swaption straddle struck at-the-money forward in the lognormal model. Setting $X = F$ and using Equations E.10–E.13, we have the value of the straddle under lognormality

$$2 \times A \times F \times \left[N\left(\frac{\sigma_{LN}\sqrt{T}}{2}\right) - N\left(-\frac{\sigma_{LN}\sqrt{T}}{2}\right)\right]. \tag{E.14}$$

We will now determine the relationship between σ_N (normal implied vol in the normal model) and σ_{LN} (lognormal implied vol in the lognormal model) for at-the-money swaptions. In order to do this, we will set the two expressions for the price of swaption straddles as given in Equations E.9 and E.14 equal to one another and deduce the restrictions that this places on the volatilities. To this end, let us first do some work to analyze the expression

$$\left[N\left(\frac{\sigma_{LN}\sqrt{T}}{2}\right) - N\left(-\frac{\sigma_{LN}\sqrt{T}}{2}\right)\right], \tag{E.15}$$

which appears as a factor in the pricing formula for the swaption straddle under lognormality in Equation E.14. Although the expression in Equation E.15 is not analytically tractable—the $N(\cdot)$ function refers to an integral that cannot be further simplified—we can use a Taylor series expansion of the cumulative normal distribution function to approximate the expression arbitrarily well.

The *error function*, denoted $\mathrm{erf}(x)$, is well known in mathematics and is defined by

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{n!(2n+1)} = \frac{2}{\sqrt{\pi}} \left(x - \frac{x^3}{3} + \frac{x^5}{10} - \frac{x^7}{42} + \cdots \right). \tag{E.16}$$

The error function and the cumulative normal distribution function are closely related by the identity

$$N(x) = \frac{1}{2}\left[1 + \mathrm{erf}\left(\frac{x}{\sqrt{2}}\right)\right]. \tag{E.17}$$

Referring now to Equation E.15, let $\epsilon = \sigma_{LN}\sqrt{T}/2$ and notice that

$$\begin{aligned} N(\epsilon) - N(-\epsilon) &= \frac{1}{2}\left[1 + \mathrm{erf}\left(\frac{\epsilon}{\sqrt{2}}\right)\right] - \frac{1}{2}\left[1 + \mathrm{erf}\left(-\frac{\epsilon}{\sqrt{2}}\right)\right] \\ &= \frac{1}{2}\left[\mathrm{erf}\left(\frac{\epsilon}{\sqrt{2}}\right) - \mathrm{erf}\left(-\frac{\epsilon}{\sqrt{2}}\right)\right] \\ &= \frac{1}{2}\left[\mathrm{erf}\left(\frac{\epsilon}{\sqrt{2}}\right) + \mathrm{erf}\left(\frac{\epsilon}{\sqrt{2}}\right)\right] \\ &= \mathrm{erf}\left(\frac{\epsilon}{\sqrt{2}}\right), \end{aligned} \tag{E.18}$$

where we have made use of the identity

$$\mathrm{erf}\left(-\frac{\epsilon}{\sqrt{2}}\right) = -\mathrm{erf}\left(\frac{\epsilon}{\sqrt{2}}\right).$$

Substituting the results of Equation E.18 into Equation E.14 gives us the price of swaption straddle under the lognormal model as

$$2 \times A \times F \times \mathrm{erf}\left(\frac{\epsilon}{\sqrt{2}}\right). \tag{E.19}$$

Equating this price with the price of the swaption straddle under the normal model implies that we must have

$$\sqrt{T} \times \sqrt{\frac{2}{\pi}} \times \sigma_N \times A = 2 \times A \times F \times \mathrm{erf}\left(\frac{\epsilon}{\sqrt{2}}\right). \tag{E.20}$$

Let $\beta = \epsilon/\sqrt{2}$, and substituting the definition of erf($\cdot$) from Equation E.16 into Equation E.20 implies that we must have

$$\sqrt{T} \times \sqrt{\frac{2}{\pi}} \times \sigma_N = 2 \times F \times \frac{2}{\sqrt{\pi}} \times \left(\beta - \frac{\beta^3}{3} + \text{error}\right), \tag{E.21}$$

where "error" denotes a (small) error term. From Equation E.21 we can write

$$\sqrt{T} \times \sqrt{\frac{2}{\pi}} \times \sigma_N \approx 2 \times F \times \frac{2}{\sqrt{\pi}} \times \left(\beta - \frac{\beta^3}{3}\right). \tag{E.22}$$

Recall that $\beta = \sigma_{LN}\sqrt{T}/(2\sqrt{2})$ and substituting into Equation E.22 gives

$$\sigma_N \approx F \times \sigma_{LN} \times \left(1 - \frac{\sigma_{LN}^2\, T}{24}\right). \tag{E.23}$$

Equation E.23 shows why we often approximate normal vol with the quantity $F \times \sigma_{LN}$: the term in parentheses is often close to one, provided that σ_{LN} and T are small. However, when σ_{LN} is large and/or time to expiration T is large, the quantity in Equation E.23 provides a meaningfully better approximation.

E.2 The Relationship Between σ_{LN} and σ_N for Off-the-Money Swaptions

The relationship between lognormal implied vol, σ_{LN}, and normal implied vol, σ_N, for swaptions with an arbitrary strike can be determined with a numerical procedure that we will outline below.[1] Before we begin, recall that under the normal model the value of a T into n receiver swaption with strike X is equal to

$$A\sigma_N\sqrt{T}\left[\frac{1}{\sqrt{2\pi}}e^{-d^2/2} - dN(-d)\right], \tag{E.24}$$

with d and A as defined in Equations E.6 and E.7, respectively.

[1] We can regard lognormal implied vol and normal implied vol as functions of X, the strike of the swaption. Denote these functions by $\sigma_{LN}(X)$ and $\sigma_N(X)$, respectively. Both Zhou (2003) and Sadr (2009) suggest that

$$\sigma_N(X) \approx \sigma_{LN}(X) \times \sqrt{FX},$$

which reduces to Equation 5.27 for the case of $X = F$ (that is, for the case in which the swaption is struck at-the-money forward). The numerical procedure developed in this section is an alternative to this approximation formula.

Sensitivity	Partial	Payer	Receiver
Delta	$\frac{\partial}{\partial F}$	$AN(d)$	$-AN(-d)$
Gamma	$\frac{\partial^2}{\partial F^2}$	$A\frac{n(d)}{\sigma_N\sqrt{T}}$	$A\frac{n(d)}{\sigma_N\sqrt{T}}$
Vega	$\frac{\partial}{\partial \sigma_N}$	$A\sqrt{T}n(d)$	$A\sqrt{T}n(d)$

Table E.1: Option Sensitivities Under the Normal Model

Recall that under the lognormal model the value of a T into n receiver swaption with strike X is given in Equation E.11 and is

$$A \times [XN(-d_2) - FN(-d_1)], \tag{E.25}$$

where d_1 and d_2 are as defined in Equations E.12 and E.13, respectively.

We assume that the normal model holds (in which case σ_N does not vary as we allow the strike of swaptions of a particular expiry and tail to vary) and seek to derive restrictions on σ_{LN}. For a given expiry T, a tail n, and a given strike X:

1. Price up the receiver swaption under the normal model for a given level of normal implied vol, σ_N, using Equation E.24. Denote by price_N the price we obtain.

2. Set the value of a T into n receiver swaption struck at X under the lognormal model in Equation E.25 equal to the price we just solved for in Step 1. That is, set

$$A \times [XN(-d_2) - FN(-d_1)] = \text{price}_N. \tag{E.26}$$

3. Determine the level of lognormal implied vol, σ_{LN}, that satisfies Equation E.26.

Following this procedure for a variety of strikes X allows us to (numerically) deduce the relationship between σ_N and σ_{LN} under the assumption of the normal model. This process was used to create Figure 5.15.[2]

[2]This procedure could be modified to utilize payer swaptions instead of receiver swaptions or even swaption straddles. The point here is to require that the value of an option is the same regardless of which model we use, then to derive restrictions on the implied vols that must be used in each model.

Sensitivity	Partial	Payer	Receiver
Delta	$\frac{\partial}{\partial F}$	$AN(d_1)$	$-AN(-d_1)$
Gamma	$\frac{\partial^2}{\partial F^2}$	$A\frac{n(d_1)}{F\sigma_{LN}\sqrt{T}}$	$A\frac{n(d_1)}{F\sigma_{LN}\sqrt{T}}$
Vega	$\frac{\partial}{\partial \sigma_{LN}}$	$AF\sqrt{T}n(d_1)$	$AF\sqrt{T}n(d_1)$

Table E.2: Option Sensitivities Under the Lognormal Model

E.3 Option Sensitivities Under the Normal Model

The option sensitivities for payer swaptions and receiver swaptions in the normal model are found by partially differentiating the expressions in Equations E.5 and E.8, respectively. Table E.1 lists some of these commonly used sensitivities. In Table E.1, d and $N(d)$ are as defined in Equations E.6 and E.4, respectively, and

$$n(d) = \frac{1}{\sqrt{2\pi}} e^{-d^2/2}. \tag{E.27}$$

By comparison, the sensitivities for payer swaptions and receiver swaptions under the lognormal (Black) model are shown in Table E.2. These sensitivities are found by partially differentiating the expressions in Equations E.10 and E.11, respectively. In Table E.2, d_1 is as given in Equation E.12.

Solutions to Selected Problems

Chapter 1

1. a) The interpolated 4-year Treasury yield is computed using the 3-year and 5-year Treasury yields. The 4-year interpolated yield is

$$0.980\% + \left(\frac{4-3}{5-3}\right) \times (1.877\% - 0.980\%) = 1.4285\%.$$

Thus, the bid side 4-year swap rate is $1.4285\% + 0.2475\% = 1.676\%$, and the offer side 4-year swap rate is $1.4285\% + 0.2525\% = 1.681\%$.

2. a) The 5-year offer side rate semi bond is equal to the bid side of the 5-year Treasury yield plus the offer side of the 5-year spread, which is $1.877\% + 0.2125\% = 2.0895\%$. This would typically be rounded to 2.090%.

3. a) The 5-year bid side swap rate is 5.25% semi bond. First, we will convert this rate to a quarterly bond basis and then to a quarterly money basis:

$$4 \times \left[\left(1 + \frac{0.0525}{2}\right)^{2/4} - 1\right] = 5.216\%$$

$$5.216\% \times 360/365 = 5.145\%.$$

The 5-year offer side swap rate is 5.27% semi bond. First, we will convert this rate to a quarterly bond basis and then to a quarterly money basis:

$$4 \times \left[\left(1 + \frac{0.0527}{2}\right)^{2/4} - 1\right] = 5.236\%$$

$$5.236\% \times 360/365 = 5.164\%.$$

Note that since the floating side of the swap remains 3-month LIBOR, there is no need to adjust for any basis swap despite the fact that the fixed side is paying quarterly.

b) The 5-year bid side swap rate is 5.25% semi bond. First, we will convert this rate to a monthly bond basis and then to a monthly money basis:

$$12 \times \left[\left(1 + \frac{0.0525}{2}\right)^{2/12} - 1\right] = 5.193\%$$

$$5.193\% \times 360/365 = 5.122\%.$$

The 5-year offer side swap rate is 5.27% semi bond. First, we will convert this rate to a monthly bond basis and then to a monthly money basis:

$$12 \times \left[\left(1 + \frac{0.0527}{2}\right)^{2/12} - 1\right] = 5.213\%$$

$$5.213\% \times 360/365 = 5.142\%.$$

c) The floating side is now 1-month LIBOR, and thus we need to incorporate the value of the basis swap. When the dealer is paying fixed and receiving 1-month LIBOR, we must use the offer side of the basis swap, which is +3 bps. Since the dealer is willing to pay 5.122% monthly money versus 3-month LIBOR, this implies that the dealer would be willing to pay $5.122\% - 0.03\% = 5.092\%$ monthly money versus 1-month LIBOR. Figure S.1 shows this calculation.

When the dealer is receiving fixed and paying 1-month LIBOR, we must use the bid side of the basis swap, which is +1 bp. Since the dealer is willing to receive 5.142% monthly money versus 3-month LIBOR, this implies that the dealer would be willing to receive $5.142\% - 0.01\% = 5.132\%$ monthly money versus 1-month LIBOR.

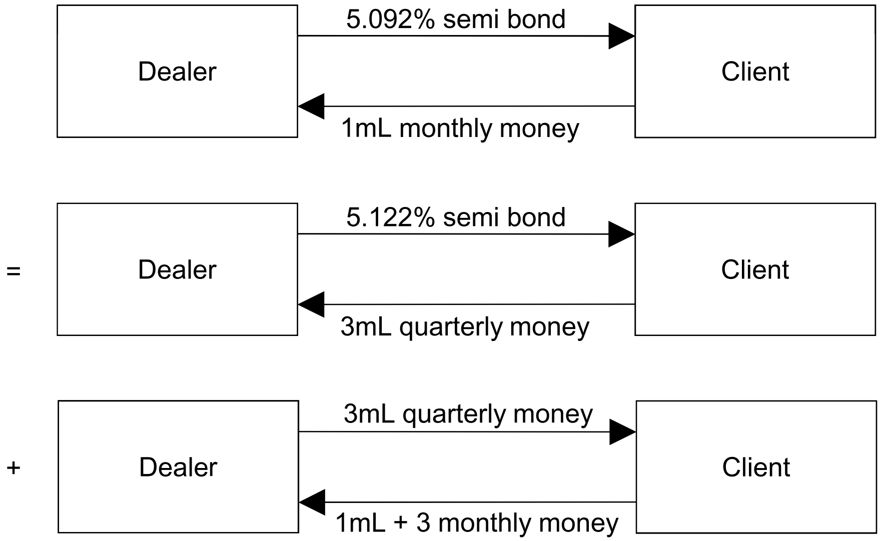

Figure S.1: Computing the Bid Side Swap Rate Versus 1-Month LIBOR

Chapter 2

1. The corporation can issue a floater at par and pay coupons of 3-month LIBOR + 66, or it can issue a fixed rate bond and pay coupons equal to $T_5 + 96$ and enter into a 5-year swap to create synthetic floating funding. If the corporation enters into the swap, it will receive fixed from the dealer at the bid side. Since mid-market swap spreads in five years are 34 bps, this means that the corporation will receive a fixed rate of $T_5 + 32$ versus LIBOR. Issuing debt at $T_5 + 96$ and receiving fixed at $T_5 + 32$ versus LIBOR nets out to paying LIBOR + 64. Figure S.2 may be helpful.

Thus, the corporation would be better off issuing the fixed rate bond and swapping to floating, saving 2 bps on the coupon. By the way, if we wanted to be very careful, we would recognize that 64 bps paid semi bond is not the same as 64 bps paid quarterly money. The equivalent number of bps paid quarterly money will be less than 64 (albeit a fraction of a bp less as determined by a proprietary swap model). This means that the corporation would obtain synthetic floating rate funding at a better level than LIBOR + 64. This realization does not change our answer; the corporation is still better off issuing fixed and swapping to floating.